Dataset 1: E-commerce

Table name: ecommerce

Operation 1: Filtering with WHERE

Question: Which orders are classified as high value?

```
SELECT OrderID, CustomerName, City, Product, TotalAmount
FROM ecommerce
WHERE OrderCategory = 'High Value';
```

What it demonstrates: Filtering records based on a business category.

Operation 2: Sorting with ORDER BY and LIMIT

Question: What are the five largest orders?

```
SELECT OrderID, CustomerName, Product, TotalAmount
FROM ecommerce
ORDER BY TotalAmount DESC
LIMIT 5;
```

What it demonstrates: Sorting order amounts from highest to lowest and returning only five rows.

Operation 3: Aggregation with GROUP BY

Question: How much order value does each city generate?

```
SELECT City,
    COUNT(*) AS NumberOfOrders,
    SUM(TotalAmount) AS TotalOrderAmount
FROM ecommerce
GROUP BY City
ORDER BY TotalOrderAmount DESC;
```

What it demonstrates: Counting orders and calculating total order amounts for each city.

Operation 4: Filtering groups with HAVING

Question: Which cities have at least two orders?

```
SELECT City,
```

```
    COUNT(*) AS NumberOfOrders,
    SUM(TotalAmount) AS TotalOrderAmount
FROM ecommerce
GROUP BY City
HAVING COUNT(*) >= 2
ORDER BY NumberOfOrders DESC;
```

What it demonstrates: HAVING filters grouped results after aggregation.

Operation 5: Subquery

Question: Which orders have a total amount greater than the average order amount?

```
SELECT OrderID, CustomerName, Product, TotalAmount
FROM ecommerce
WHERE TotalAmount > (
  SELECT AVG(TotalAmount)
  FROM ecommerce
)
ORDER BY TotalAmount DESC;
```

What it demonstrates: Comparing individual orders against a value calculated by a nested query.

Operation 6: Date-based analysis

Question: What is the total order amount for each order date?

```
SELECT OrderDate,
    COUNT(*) AS NumberOfOrders,
    SUM(TotalAmount) AS DailyOrderAmount
FROM ecommerce
GROUP BY OrderDate
ORDER BY OrderDate;
```

What it demonstrates: Grouping transactions by date to identify daily order patterns.

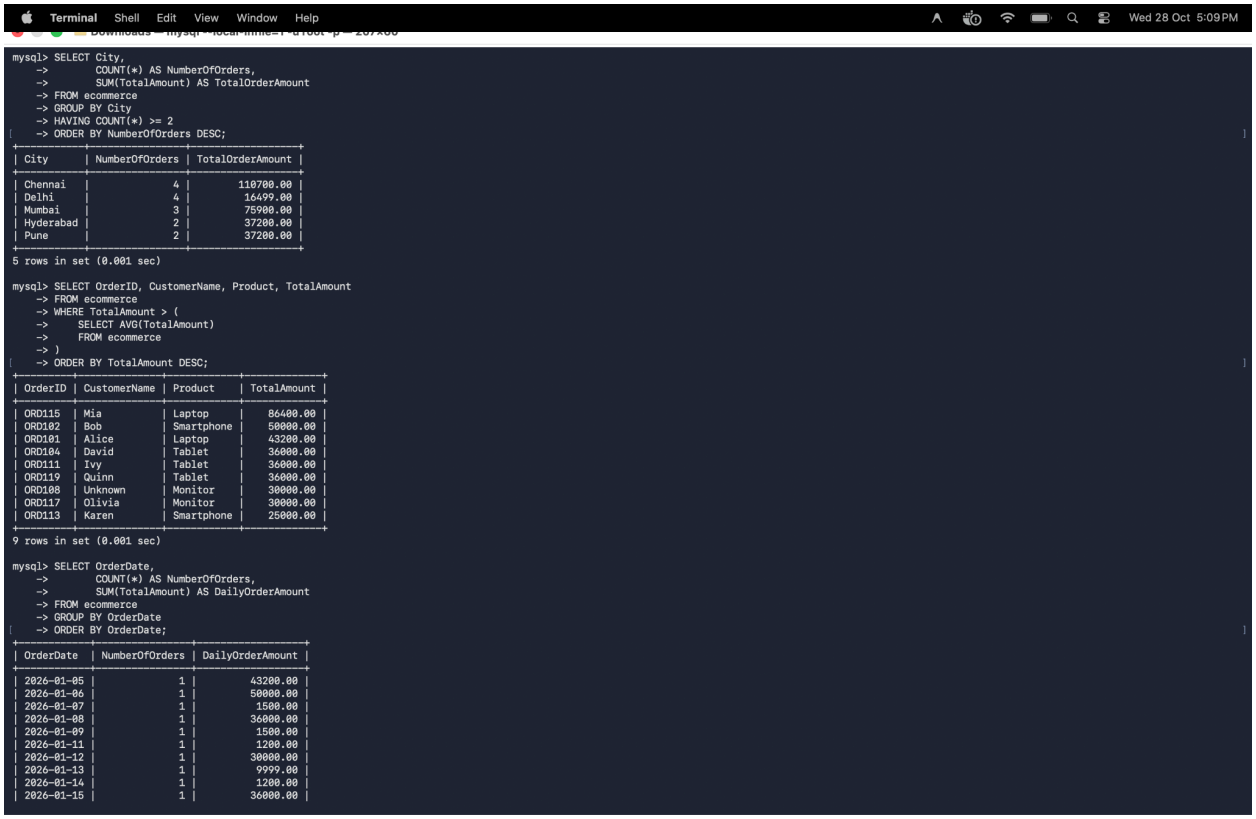
Operation 7: Duplicate detection

Question: Are any order IDs repeated?

```
SELECT OrderID,
    COUNT(*) AS Occurrences
FROM ecommerce
GROUP BY OrderID
HAVING COUNT(*) > 1;
```

What it demonstrates: Identifying repeated order IDs. If the result is empty, no duplicate IDs were found

Results



```
mysql> SELECT OrderID, CustomerName, Product, TotalAmount
    -> FROM ecommerce
    -> WHERE TotalAmount > (
    ->     SELECT AVG(TotalAmount)
    ->     FROM ecommerce
    -> )
    -> ORDER BY TotalAmount DESC;
+---------+--------------+------------+-------------+
| OrderID | CustomerName | Product    | TotalAmount |
+---------+--------------+------------+-------------+
| ORD115  | Mia          | Laptop     |    86400.00 |
| ORD102  | Bob          | Smartphone |    50000.00 |
| ORD101  | Alice        | Laptop     |    43200.00 |
| ORD104  | David        | Tablet     |    36000.00 |
| ORD111  | Ivy          | Tablet     |    36000.00 |
| ORD119  | Quinn        | Tablet     |    36000.00 |
| ORD108  | Unknown      | Monitor    |    30000.00 |
| ORD117  | Olivia       | Monitor    |    30000.00 |
| ORD113  | Karen        | Smartphone |    25000.00 |
+---------+--------------+------------+-------------+
9 rows in set (0.001 sec)

mysql> SELECT OrderDate,
    ->        COUNT(*) AS NumberOfOrders,
    ->        SUM(TotalAmount) AS DailyOrderAmount
    -> FROM ecommerce
    -> GROUP BY OrderDate
    -> ORDER BY OrderDate;
+------------+-----------------+------------------+
| OrderDate  | NumberOfOrders | DailyOrderAmount |
+------------+-----------------+------------------+
| 2026-01-05 |               1 |         43200.00 |
| 2026-01-06 |               1 |         50000.00 |
| 2026-01-07 |               1 |          1500.00 |
| 2026-01-08 |               1 |         36000.00 |
| 2026-01-09 |               1 |          1500.00 |
| 2026-01-11 |               1 |          1200.00 |
| 2026-01-12 |               1 |         30000.00 |
| 2026-01-13 |               1 |          9999.00 |
| 2026-01-14 |               1 |          1200.00 |
| 2026-01-15 |               1 |         36000.00 |
| 2026-01-17 |               1 |         25000.00 |
| 2026-01-18 |               1 |          4500.00 |
| 2026-01-19 |               1 |         86400.00 |
| 2026-01-20 |               1 |          1200.00 |
| 2026-01-21 |               1 |         30000.00 |
| 2026-01-22 |               1 |           900.00 |
| 2026-01-23 |               1 |         36000.00 |
| 2026-01-24 |               1 |           500.00 |
+------------+-----------------+------------------+
18 rows in set (0.001 sec)

mysql> SELECT OrderID,
    ->        COUNT(*) AS Occurrences
    -> FROM ecommerce
    -> GROUP BY OrderID
    -> HAVING COUNT(*) > 1;
Empty set (0.001 sec)

mysql>
```

```
mysql> SELECT * FROM ecommerce LIMIT 5;
+---------+--------------+-----------+------------+----------+-----------+------------+---------------+---------------------+-------------+---------------+---------+
| OrderID | CustomerName | City      | Product    | Quantity | UnitPrice | OrderDate  | PaymentMethod | UnitPrice_Original  | TotalAmount | OrderCategory | Revenue |
+---------+--------------+-----------+------------+----------+-----------+------------+---------------+---------------------+-------------+---------------+---------+
| ORD101  | Alice        | Chennai   | Laptop     |        1 |  43200.00 | 2026-01-05 | Credit Card   |            75000.00 |    43200.00 | Medium Value  | 75000.00 |
| ORD102  | Bob          | Mumbai    | Smartphone |        2 |  25000.00 | 2026-01-06 | Upi           |            25000.00 |    50000.00 | High Value    | 50000.00 |
| ORD103  | Charlie      | Delhi     | Headphones |        1 |   1500.00 | 2026-01-07 | Cash          |             1500.00 |     1500.00 | Low Value     |  1500.00 |
| ORD104  | David        | Hyderabad | Tablet     |        2 |  18000.00 | 2026-01-08 | Debit Card    |            18000.00 |    36000.00 | Medium Value  | 36000.00 |
| ORD105  | Eve          | Chennai   | Mouse      |        3 |    500.00 | 2026-01-09 | Upi           |              500.00 |     1500.00 | Low Value     |  1500.00 |
+---------+--------------+-----------+------------+----------+-----------+------------+---------------+---------------------+-------------+---------------+---------+
5 rows in set (0.001 sec)

mysql> SELECT OrderID, CustomerName, City, Product, TotalAmount
    -> FROM ecommerce
    -> WHERE OrderCategory = 'High Value';
+---------+--------------+--------+------------+-------------+
| OrderID | CustomerName | City   | Product    | TotalAmount |
+---------+--------------+--------+------------+-------------+
| ORD102  | Bob          | Mumbai | Smartphone |    50000.00 |
| ORD115  | Mia          | Kolkata| Laptop     |    86400.00 |
+---------+--------------+--------+------------+-------------+
2 rows in set (0.001 sec)

mysql> SELECT OrderID, CustomerName, Product, TotalAmount
    -> FROM ecommerce
    -> ORDER BY TotalAmount DESC
    -> LIMIT 5;
+---------+--------------+------------+-------------+
| OrderID | CustomerName | Product    | TotalAmount |
+---------+--------------+------------+-------------+
| ORD115  | Mia          | Laptop     |    86400.00 |
| ORD102  | Bob          | Smartphone |    50000.00 |
| ORD101  | Alice        | Laptop     |    43200.00 |
| ORD104  | David        | Tablet     |    36000.00 |
| ORD111  | Ivy          | Tablet     |    36000.00 |
+---------+--------------+------------+-------------+
5 rows in set (0.001 sec)

mysql> SELECT City,
    ->        COUNT(*) AS NumberOfOrders,
    ->        SUM(TotalAmount) AS TotalOrderAmount
    -> FROM ecommerce
    -> GROUP BY City
    -> ORDER BY TotalOrderAmount DESC;
+-----------+-----------------+------------------+
| City      | NumberOfOrders | TotalOrderAmount |
+-----------+-----------------+------------------+
| Chennai   |               4 |        110700.00 |
| Kolkata   |               1 |         86400.00 |
| Mumbai    |               3 |         75900.00 |
| Hyderabad |               2 |         37200.00 |
| Pune      |               2 |         37200.00 |
| Unknown   |               1 |         30000.00 |
| Delhi     |               4 |         16499.00 |
| Bengaluru |               1 |          1200.00 |
+-----------+-----------------+------------------+
8 rows in set (0.002 sec)

mysql> SELECT City,
```